Homework 3 Solutions

Problem 1

Given the provided dataset "HW3_data.csv", consider the model regressing y on x and m. Conduct a Sobel test to see whether there is a mediation effect. You need to consider the three relevant models,

$$y_i = \beta_{11} + \beta_{12}x_i + \epsilon_i,$$

$$m_i = \beta_{21} + \beta_{22}x_i + \epsilon_i,$$

and

$$y_i = \beta_{31} + \beta_{32}m_i + \beta_{33}x_i + \epsilon_i.$$

Suppose the variance of ϵ_i is known with $\sigma^2 = 1$ for all the three models. Use the level of significance $\alpha = 0.05$

(1) (4 pts) What is the null hypothesis of the Sobel test?

Solution:

The null hypothesis is that there's no mediation effect, $H_0: \ \beta_{12} = \beta_{33}$

(2) (8 pts) Regress the three models and find the estimators for $(\beta_{11},\ \beta_{12})$, $(\beta_{21},\ \beta_{22})$ and $(\beta_{31},\ \beta_{32},\ \beta_{33})$.

Solution:

$$(\hat\beta_{11},\ \hat\beta_{12}) = (-1.607,\ 2.638), \qquad (\hat\beta_{21},\ \hat\beta_{22}) = (-0.772,\ 2.055)$$

$$(\hat\beta_{31},\ \hat\beta_{32},\ \hat\beta_{33}) = (-0.666,\ 1.219,\ 0.133)$$

(3) (10 pts) Find the variance of $\hat\beta_{22}$ and $\hat\beta_{32}$.

Solution: Recall that, for a linear model $Y = X\beta + \sigma\epsilon$, $Var(\hat\beta_j) = \sigma^2 C_j$, where C_j is the jth diagonal element of $(X'X)^{-1}$.

$$Var(\hat\beta_{22}) = 0.0011 \qquad Var(\hat\beta_{32}) = 0.0084$$

(4) (10 pts) Find the z test statistic.

Solution:

$$z = \frac{\hat\beta_{12} - \hat\beta_{33}}{\sqrt{\hat\beta_{22}^2 Var(\hat\beta_{32}) + \hat\beta_{32}^2 Var(\hat\beta_{22})}} = 12.995$$

(5) (8 pts) Find the p-value and the conclusion of the test.

Solution:

$$p = 2(1 - \text{pnorm}(z, 0, 1)) = 0 < \alpha = 0.05$$

So, we reject the null hypothesis of no mediation effect.

Problem 2

Consider the usual simple linear model (i.e. there is only a single covariate and hence only a single unknown β, assuming the data is centered, so no intercept term),

$$y_i = x_i\beta + \sigma\epsilon_i,$$

where σ is unknown and $\sum_{i=1}^n x_i^2 = n$.

(1) (12 pts) Show that the least square estimator is given by $\widehat{\beta} = n^{-1}\sum_{i=1}^n x_i y_i$.

Solution:

$$I(\beta) = \sum_{i=1}^n (y_i - x_i\beta)^2 = \sum_{i=1}^n (y_i^2 + x_i^2\beta^2 - 2x_i y_i \beta)$$

$$\frac{dI(\beta)}{d\beta} = \sum_{i=1}^n (2x_i^2\beta - 2x_i y_i) = 2\beta\sum_{i=1}^n x_i^2 - 2\sum_{i=1}^n x_i y_i = 2n\beta - 2\sum_{i=1}^n x_i y_i = 0$$

$$\text{So} \quad \hat{\beta} = n^{-1}\sum_{i=1}^n x_i y_i$$

(2) (12 pts) Show that the residual sum of squares is given by $\sum_{i=1}^n y_i^2 - n\,\widehat{\beta}^2$.

Solution:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - x_i\hat{\beta})^2 = \sum_{i=1}^n y_i^2 - 2\hat{\beta}\sum_{i=1}^n x_i y_i + \hat{\beta}^2\sum_{i=1}^n x_i^2$$

$$= \sum_{i=1}^n y_i^2 - 2\hat{\beta}n\hat{\beta} + \hat{\beta}^2 n = \sum_{i=1}^n y_i^2 - n\hat{\beta}^2$$

(3) (12 pts) Find the F statistic for testing $H_0 : \beta = 0$

Solution:

Here the reduced model is

$$y_i = \sigma\epsilon_i \qquad \hat{y}_{red} = 0 \qquad \hat{e}_{red}'\hat{e}_{red} = \sum_{i=1}^n y_i^2$$

$$\hat{e}'\hat{e} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n y_i^2 - n\hat{\beta}^2$$

$$F = \frac{\hat{e}_{red}'\hat{e}_{red} - \hat{e}'\hat{e}}{\hat{e}'\hat{e}/(n-p)} = \frac{\sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i^2 - n\hat{\beta}^2)}{(\sum_{i=1}^n y_i^2 - n\hat{\beta}^2)/(n-1)} = \frac{n\hat{\beta}^2(n-1)}{\sum_{i=1}^n y_i^2 - n\hat{\beta}^2}$$

(4) (12 pts) If the null hypothesis is true, what is the distribution of $\widehat{\beta}$?

Solution:

If the null hypothesis $H_0 : \beta = 0$ is true, than

$$y_i = \sigma\epsilon_i \qquad \hat{\beta} = n^{-1}\sum_{i=1}^{n} x_i y_i = n^{-1}\sigma\sum_{i=1}^{n} x_i\epsilon_i \qquad \epsilon_i \overset{iid}{\sim} N(0,1)$$

$\hat{\beta}$ is a linear combination of n independent normal random variables, so it's also normally distributed, with mean and variance,

$$E(\hat{\beta}) = n^{-1}\sigma\sum_{i=1}^{n} x_i E(\epsilon_i) = 0 \qquad Var(\hat{\beta}) = n^{-2}\sigma^2\sum_{i=1}^{n} x_i^2 Var(\epsilon_i) = \sigma^2/n$$

so $\hat{\beta} \sim N(0,\ \sigma^2/n)$.

(5) (12 pts) Hence, show that the Student–t test statistic for testing the null hypothesis is

$$T = \sqrt{n}\widehat{\beta}\,/\,\widehat{\sigma}, \qquad \text{with} \quad \widehat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

and confirm that the F test and the T test are the same.

Solution:

$$\frac{\sqrt{n}\hat{\beta}}{\sigma} \sim N(0,1) \qquad (n-1)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$\text{So} \qquad T = \frac{\sqrt{n}\hat{\beta}}{\hat{\sigma}} = \frac{\sqrt{n}\hat{\beta}/\sigma}{\sqrt{(n-1)\frac{\hat{\sigma}^2}{\sigma^2}/(n-1)}} = \frac{N(0,\ 1)}{\sqrt{\chi^2_{n-1}/(n-1)}} \sim t_{n-1}$$

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} y_i^2 - n\hat{\beta}^2\right).$$

Hence,

$$T^2 = \frac{n\hat{\beta}^2}{\hat{\sigma}^2} = \frac{n\hat{\beta}^2(n-1)}{\sum_{i=1}^{n} y_i^2 - n\hat{\beta}^2} = F$$